

Monitoring Valence with Facial EMG Sensors using Machine Learning

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ABSTRACT

Affective Computing is an essential field of study with many impactful applications, from remote solutions for timely detection and improved management of mental health to improved human-computer interaction in the physical world and the metaverse. In this study, we explore the usage of a novel facial mask equipped with seven surface electromyography (sEMG) sensors to monitor subjective valence. We collected data from 30 participants who watched 20 affective videos (five exciting videos, five positive videos, five neutral videos, and five negative videos). The sensor data was filtered, and EMG features were extracted, including statistical and frequency-based features. The exploratory analysis indicated that the activations of the muscles associated with positive affect (left/right orbicularis and left/right zygomaticus) increased during positive periods compared to negative ones. Similarly, the activation of the corrugator muscle (related to negative affect) was increased during negative periods. This trend was also confirmed by the SHAP analysis performed on a machine-learning model for classifying 3-class valence (negative, neutral, and positive). On the test data of five unseen participants, the model achieved an accuracy of 63% and an F1-score of 49%. The class imbalance seemed to be one of the more significant challenges the ML models faced. The results confirmed the relationship between sEMG sensing and subjective valence but also showed that monitoring valence in real-time is challenging and requires more advanced approaches.

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CCS CONCEPTS

• **Applied computing**; • **Life and medical sciences**; • **Health informatics**;

KEYWORDS

emotional valence, affect, machine learning, classification, wearable sensors, facial EMG

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1 INTRODUCTION

The emotional states that we experience in our everyday interactions with the world around us and different types of media are conveyed by complex and subtle facial gestures. By applying EMG sensors to the regions of the face with the most prominent muscle activity, we can harness this information and use it to detect the user's affect (i.e., their current mood or emotion). Virtual reality has been a growing trend in the modern world of media consumption, so it is only natural to seek new ways to improve the user experience [1]. Besides the use-cases in human-computer interaction, remote solutions for timely detection and improved management of emotion disorders will positively impact the lives of over one hundred million people in the EU alone who experience mental health problems [2]. This situation was further aggravated by the COVID-19 pandemic, which increased mental health problems' prevalence [3].

Many theories have delved into defining the concept of emotions and affect. For this study, we will use the dimensional model of

affect (Russell, 1980 [4]), which states that affect is composed of valence and arousal. Valence refers to the positive and negative scale that describes the emotion, whereas arousal refers to the intensity of the emotion. This study focuses on monitoring valence because of its connection to facial muscle activation.

This study aims to explore the usage of a novel facial mask equipped with surface electromyography (sEMG) sensors to monitor the user's subjective valence. Our approach is based on signal-processing and machine learning (ML) techniques to develop valence classification models. While EMG-based analysis has been used for decades to study the complexities of human muscles and their relations to affect [5], it was only recently that less obtrusive sEMG-based wearables became available on the market [6]. Furthermore, many studies have been conducted to analyze users' facial expressions based on data collected from sEMG sensors. However, most of them focus on predicting a specific emotion or facial expression rather than valence. For example, Kelati et al. [8] used a support vector machine (SVM) model to detect facial expressions (smiling & frowning) from sEMG data. Mithbavkar and Shah [9], and Cheng and Liu [10] conducted similar studies, focusing on predicting emotional states (joy, anger, sadness, and pleasure) using neural networks and EMG data. Similarly, Yadav et al. [11] developed classification models for valence using EMG data. It should be noted that our study uses data from a wearable face mask equipped with sEMG sensors. sEMG data is less obtrusive to collect than EMG data. However, it may also bring more challenges, e.g., noise caused by movements, which is less problematic for EMG-based analysis, where the contact between the skin and EMG electrodes is better secured.

2 DATA

We recorded data from 30 healthy participants aged 16 to 23 using the emteq's emteqPROtm face mask [7] [12]. The face mask has seven EMG sensors (Figure 1): two frontalis sensors (6 and 0) used to monitor eyebrow movement; two orbicularis sensors (4 and 2) used to monitor eye movements; two zygomaticus sensors (5 and 1) used to monitor mouth and cheek movements; one corrugator sensor (3) used to monitor forehead movements. The mask can be used as an add-on to standard VR headsets, but it can also be used in an open-face manner (e.g., while working on a PC) without a VR headset. We used the open-face setup in our experiments.

In the beginning, we recorded a baseline session for each participant, during which the participants watched a neutral 2-minute video and were instructed to be as relaxed and neutral as possible. After the baseline session, the participants watched 20 videos on a laptop. On a scale of 1 to 9, the participants rated the level of emotional valence they felt during the video. The videos were divided into four groups: exciting videos, positive videos, neutral videos, and negative videos. Each group had five videos. The subjective valence ratings are shown in Figure 2. The figure shows that the subjective ratings are related to the four pre-defined video types. The average rating for the negative videos (red) is around three. The average rating for the positive videos (green) is between 6 and 7. The average rating for the neutral videos (yellow) is around 5, with two videos being more positive. The average ratings for the exciting videos span from 3 to 6, which signifies that some exciting



Figure 1: emteqPROtm multi-sensor face mask.

videos can be positively valenced and some exciting videos can be negatively valenced.

3 APPROACH

3.1 Preprocessing and feature extraction

The sEMG signals provided by the mask are sampled at a 1000 Hz sampling frequency. In the beginning, we performed signal denoising and filtering. The EMG signals were initially filtered with a Hampel filter to remove abrupt peaks in the signals that appear because of rapid movements. Additionally, we used a frequency-based filtering method to remove electromagnetic noise at 50 Hz and its harmonics. The filtered data was then segmented, and features were extracted from each segment. The data was segmented using a 0.5-second window and a 0.1-second stride (meaning that there was an 80% overlap between consecutive windows). Eventually, we extracted 34 statistical features per EMG channel, resulting in a total of 238 features. The features included various amplitude-based features (e.g., average amplitude change and mean absolute value), amplitude derivatives, auto-regressive coefficients, cepstral coefficients, frequency-based features (e.g., main frequency) and statistical features (e.g., statistical moments). Finally, person-specific baseline normalization was performed for each subject separately by subtracting the baseline-mean value from each feature.

3.2 Modeling

The extracted features were used as input to several ML algorithms, including Random Forest (an ensemble of 100 decision trees), KNN (predicts the majority class of an instance's k nearest neighbors), XGB (gradient boosting algorithm which trains decision tree models sequentially, and each subsequent model aims to correct the errors of its predecessors), and LGB (similar to XGB but uses an additional sampling technique). For each algorithm, we tried classification- and regression-based approaches. The regressors were trained to predict the valence score between 1 and 9, and the classifiers were trained to predict three classes, negative valence (subjective scores between 1 and 3), neutral valence (subjective scores between 4 and 6), and positive valence (subjective scores between 7 and 9). In the end, the regressors' output was mapped to the same three classes, and all models were evaluated using accuracy and a macro F1-score. We were interested in person-independent models. Thus, from the

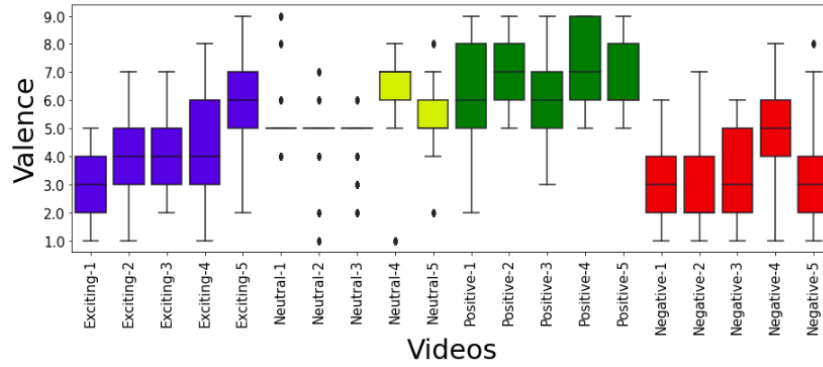


Figure 2: Subjective valence (one boxplot per video) and video types (color coded).

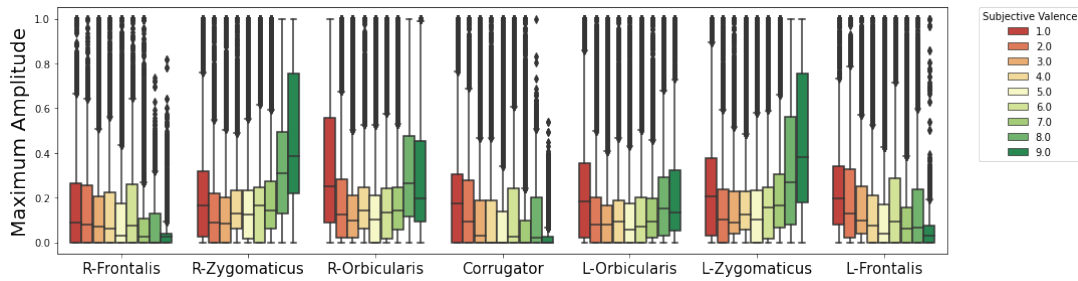


Figure 3: Boxplots representing the distribution of the maximum sEMG amplitudes for the seven EMG sensors, grouped by subjective valence ratings from one (negative valence) to nine (positive valence).

beginning, the data was split into a training (20 randomly selected participants), a validation (5 randomly selected participants), and a test set (5 randomly selected participants).

4 EXPERIMENTS

Before the ML experiments, we performed exploratory data analysis to inspect the relationship between the sensor data and the subjective valence. Figure 3 presents the distribution of the feature “maximum amplitude” via boxplots. The x-axis represents the seven sEMG sensors. Within each sEMG sensor, the data are also grouped per subjective valence level, from one (negative valence) to nine (positive valence). The valence levels are also color-coded. The boxplots of the *frontalis* muscles and the *corrugator* muscle (muscles related to frowning, squeezed eyes, and focus) show that the amplitude is highest for the videos ranked with negative valence levels (valence levels from one to four), and it decreases for the videos ranked with positive valence levels. For the videos ranked with valence level 5 (neutral valence), the amplitude of the *corrugator* sensor stays relatively high, which may be related to the participants being focused on the videos. On the other hand, for the positive valence ratings (6 and above), we can see that the activation of the *zygomaticus* muscles (muscles related to smiling expressions measured via the cheeks) is relatively higher than the activation of all the other muscles. We can also see that the *orbicularis* muscles are activated both for the negative and for the positive videos, but the activation is lower for the neutral videos. This U-shaped (or

V-shaped) activation has already been observed in previous facial EMG studies related to affect [13].

Table 1 shows the results of the ML experiments. It can be seen that, in general, the classifiers performed better than the corresponding regressors. Nevertheless, these results may also be slightly biased because of the usage of classification-based evaluation metrics. Furthermore, the LGB classifier achieved the best results, with 68% accuracy and a 50% F1-Score on the validation data and 63% accuracy and a 49% F1 score on the test data.

Table 2 shows the confusion matrix for the LGB classifier on the test data, where the rows represent the actual label, and the columns represent the predicted label. The matrix shows that the dataset is imbalanced (around 5K positive and 5K negative instances vs. 23K neutral instances). Because of the imbalance, the model often mixes positive and negative instances with the neutral class. It is encouraging to see that not many positive instances are predicted as negative.

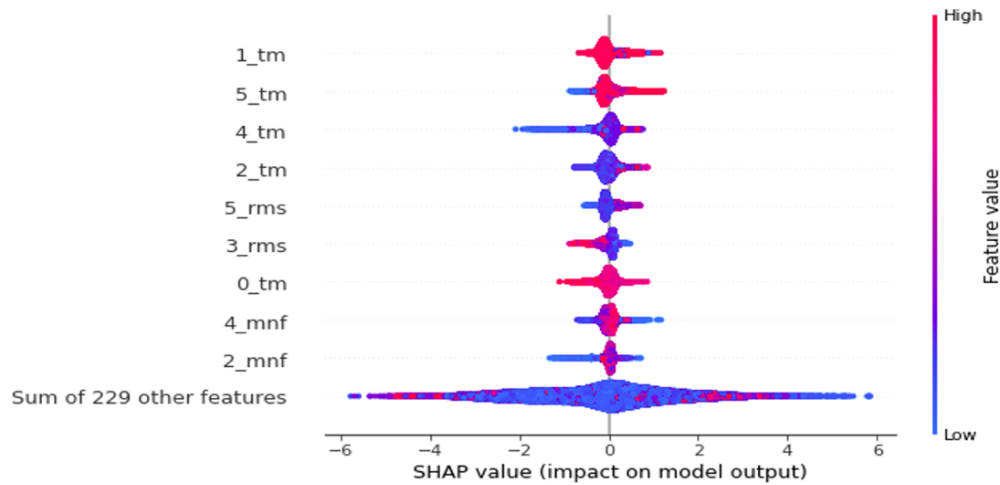
Figure 4 depicts the SHAP analysis for the best-performing model (LGB classifier). The top feature is the temporal moment (*tm*) extracted from 4 different sensors (1_tm and 5_tm – the zygomaticus sensors; 2_tm and 4_tm – the orbicularis sensors). The *tm* feature is related to the absolute amplitude of the corresponding sensor. Thus, the top-2 features (1_tm and 5_tm) show that a high *tm* value is related to positive valence (probably because of smiling expressions). For the feature 5_tm it can also be seen that low values are related to negative valence. A similar relation can be seen for 1_tm

Table 1: Results of the machine-learning experiments.

Algorithm	Classifiers				Regressors			
	Accuracy		F1 score		Accuracy		F1 score	
	Validation	Test	Validation	Test	Validation	Test	Validation	Test
RF	64%	56%	49%	46%	66%	61%	42%	42%
XGB	59%	49%	46%	43%	59%	50%	40%	37%
LGB	68%	63%	50%	49%	67%	61%	42%	36%
KNN	66%	61%	47%	43%	66%	62%	40%	38%

Table 2: Confusion matrix for the best performing model.

	Negative	Neutral	Positive
Negative	1217	3072	1263
Neutral	948	19785	2497
Positive	208	2926	2104

**Figure 4: SHAP analysis of the best performing model (LGB classifier).**

and 5_tm , i.e., a low tm value (depicting low muscle activation) is related to negative valence. Another interesting observation is the feature 3_rms , which is the root mean square of the corrugator sensor. Higher values of this feature represent higher activation of the corrugator muscle, e.g., during frowning expressions. And indeed, according to the model, higher 3_rms values are related to negative valence.

5 CONCLUSION

In this study, we investigated the association between facial sEMG, and subjective valence using a novel wearable sEMG sensing device. We analyzed sensor data from 30 participants that watched 20 affective videos (five exciting videos, five positive videos, five neutral videos, and five negative videos). The exploratory analysis (see Figure 3) indicated that the sEMG amplitude was related to the subjective valence ratings. A U-shaped (or V-shaped) activation of some muscles was observed, which aligns with existing EMG

studies on affect [12]. Furthermore, the activations of the muscles associated with positive affect (left/right orbicularis and left/right zygomaticus) increased during positive periods compared to negative ones. Similarly, the activation of the corrugator muscle (related to negative affect) was increased during negative periods. This trend was also confirmed by the SHAP analysis presented in Figure 4. The SHAP analysis was performed on the best-performing model for classifying 3-class valence (negative, neutral, and positive). On the test data of unseen participants, the model achieved an accuracy of 63% and an F1-score of 49%. The class imbalance seemed to be one of the more significant challenges the ML models faced. In the future, we plan to investigate hyperparameter optimization, feature selection, and supervised and unsupervised personalization to improve model performance. Finally, ethics, fairness, model explainability, and accountability must be explored in the context of this and similar affect recognition approaches.

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